

AstroManager

Professional Astrophotography File Management Suite

User Manual — Version 1.0

AstroManager Development Team

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1 Introduction

1.1 Welcome to AstroManager

AstroManager is a comprehensive, professional-grade file management suite designed specifically for astrophotographers. Whether you're managing hundreds or hundreds of thousands of FITS and XISF files, AstroManager provides the tools you need to organize, compress, analyze, and optimize your imaging library.

1.2 Key Features

- **Universal File Analysis:** Recursive analysis of FITS, XISF, and FITS.FZ files with detailed statistics, target grouping, and professional report generation (LaTeX/PDF, HTML, CSV, AstroBin)
- **Multi-Codec Compression:** Bidirectional conversion between formats with 9 compression profiles (zlib, zstd, lz4), saving 40–60% disk space
- **Mass Header Editing:** Edit FITS headers across thousands of files with NINA compatibility and pattern builders
- **Flat Frame Management:** Automatic grouping, coverage reports, master flat tracking, and target linking
- **Target Tracking:** Historical observation timelines with integration time statistics, best session identification, weather forecast, and target visibility
- **Plate Solving:** ASTAP/Astrometry.net integration for automatic focal reducer detection
- **Weather Integration:** Historical weather data enrichment via Open-Meteo (free, no API key)
- **Storage Optimization:** Disk space analysis, duplicate detection, file organization with pre-sets, and optimization recommendations
- **Performance:** Handles 200,000+ files efficiently with parallel processing, multi-tier caching, and auto-tuning
- **Observation History:** Comprehensive dashboard with target rankings, filter/equipment/temporal statistics, best nights, and full JSON/CSV export/import
- **Database Browser:** Searchable databases of 1,600+ cameras, 3,900+ telescopes, 1,600+ filters, and 32,900+ astronomical targets
- **Modern UI:** Cosmic-themed interface with 8 organized tabs, bilingual support (EN/FR), tooltips on every widget, and built-in settings dialog
- **Auto-Setup:** Automatic dependency installation, virtual environment creation, and system detection
- **Anonymous Bug Reporting:** Optional crash reports to help improve AstroManager

1.3 System Requirements

Optional Components:

- **LaTeX (pdflatex):** For compiled PDF reports (HTML always available as fallback)
- **ASTAP:** For plate solving and focal reducer detection
- **Internet:** For SIMBAD resolution, weather data, and bug reporting

Component	Requirement
Operating System	Windows 10/11, Linux (any distro incl. Manjaro/Arch), macOS 10.14+
Python Version	3.8 or higher
RAM	4 GB minimum, 8 GB+ recommended for 100k+ files
Disk Space	500 MB for application + space for database
Display	1280×720 minimum, 1920×1080 recommended

Table 1: System Requirements

2 Installation

2.1 Quick Installation (Recommended)

AstroManager provides fully automated launchers that handle everything: Python detection/installation, virtual environment creation, dependency installation, and optional LaTeX setup.

2.1.1 Windows

1. Download or clone the repository:

```
git clone https://github.com/ARP273-ROSE/astromanager.git
cd astromanager
```

2. Double-click `run.bat` or run from command line:

```
run.bat
```

3. The launcher will automatically:
 - Detect Python 3.8+ (install via `winget` if missing)
 - Install `winget` if needed (4 fallback methods)
 - Create a virtual environment (`venv/`)
 - Install all Python dependencies from `requirements.txt`
 - Detect and offer to install MiKTeX (LaTeX) for PDF reports
 - Launch AstroManager

2.1.2 Linux / macOS

1. Clone the repository:

```
git clone https://github.com/ARP273-ROSE/astromanager.git
cd astromanager
chmod +x run.sh
```

2. Run the launcher:

```
./run.sh
```

3. The launcher supports: `pacman` (Arch/Manjaro), `apt` (Debian/Ubuntu), `dnf` (Fedora), `zypper` (openSUSE), and `brew` (macOS).

Note: Bilingual Launchers

Both `run.bat` and `run.sh` auto-detect your system language and display all messages in English or French accordingly.

2.2 Manual Installation

If you prefer manual control:

```
# Create virtual environment (recommended)
python -m venv venv
source venv/bin/activate      # Linux/macOS
venv\Scripts\activate        # Windows

# Install all dependencies
pip install -r requirements.txt

# Launch application
python astromanager.py
```

Core dependencies: PyQt6, numpy, pandas, matplotlib, astropy, Pillow, reportlab, tqdm, requests, xisf, psutil, PyYAML

Optional: astroquery (SIMBAD), scipy (advanced statistics), zstandard (zstd compression), lz4 (lz4 compression)

2.3 Auto-Dependency Installation

On first launch, AstroManager checks for all required packages and installs any missing ones automatically using `pip`. This ensures the application works even if `requirements.txt` was not used.

2.4 First Launch

On first launch, AstroManager will:

1. Create configuration directory: `~/.astromanager/`
2. Generate default configuration: `config.yaml`
3. Initialize SQLite database: `astromanager.db`
4. Auto-detect system capabilities (CPU cores, RAM, storage type)
5. Set optimal worker count and batch size

Note: The database is automatically backed up on each launch to `astromanager_backup.db`.

3 User Interface Overview

3.1 Main Window

AstroManager features a modern, cosmic-themed interface organized into 8 functional tabs:

1. **Analysis:** File analysis and report generation
2. **Compression:** Multi-format compression and conversion
3. **Header Editor:** Mass FITS header editing
4. **Flat Manager:** Flat frame organization and management
5. **Target Tracking:** Historical observation tracking with weather forecast

6. **History & Statistics:** Comprehensive observation dashboard with rankings, stats, and export
7. **Database Browser:** Searchable camera, telescope, filter, and target databases
8. **Disk Space:** Storage analysis, optimization, and file organization

3.2 Menu Bar

- **File:** Exit (Ctrl+Q)
- **View:** Toggle console (Ctrl+‘), Language selector (English/Français)
- **Tools:** Settings (Ctrl+,), Optimize Database, Clear Old Cache, Reset Configuration
- **Help:** User Guide (F1), Keyboard Shortcuts, System Info, Report Bug, About

3.3 Console and Progress

- **Console Output:** Bottom panel showing real-time log messages (toggle with Ctrl+‘)
- **Progress Bar:** Real-time progress with phase information and ETA
- **Status Bar:** Worker count, RAM info, and current status

3.4 Tooltips

Every widget in AstroManager has a bilingual tooltip that appears on hover, explaining the function of each button, checkbox, dropdown, and input field. This makes the interface self-documenting.

4 Analysis Tab

4.1 Overview

The Analysis tab provides comprehensive FITS/XISF file analysis with automatic target detection, filter grouping, telescope identification, and professional report generation.

4.2 Basic Workflow

1. **Select Folder:** Click **Browse** and navigate to your FITS/XISF folder
2. **Configure Options:** Enable desired analysis features (SIMBAD, duplicates, plate solve, weather)
3. **Choose Output Formats:** Select LaTeX/PDF, HTML, CSV, or AstroBin export
4. **Set Workers:** Leave at 0 for automatic optimization, or set manually
5. **Start Analysis:** Click **Start Analysis** button
6. **View Results:** Switch to Results tab or open output folder

4.3 Analysis Options

4.3.1 SIMBAD Resolution

Resolves target names via the SIMBAD astronomical database:

- Merges duplicates (e.g., M31 = NGC 224 = Andromeda Galaxy)
- Provides canonical names and object types
- Fetches coordinates (RA/DEC) and magnitude
- Rate-limited to avoid server overload (0.6s between queries)
- Handles telescope aliasing and merging (e.g., “FSQ85” = “FSQ-85EDP”)

4.3.2 Duplicate Detection

Detects duplicate files using three methods:

1. **Name-based:** Same base name, different extensions (e.g., image.fits vs image.xisf)
2. **Content-based:** Identical FITS headers (DATE-OBS, EXPTIME, FILTER, etc.) — uses parallelized signature reading for speed
3. **Compressed:** Original and compressed versions (FITS + FITS.FZ pairs)

4.3.3 Plate Solving

(Optional)

Requires ASTAP or Astrometry.net installation:

- Solves plate for WCS solution
- Calculates actual image scale (arcsec/pixel)
- Compares with expected scale from telescope/camera databases
- Detects focal reducers (0.67x, 0.72x, 0.8x) with 3% tolerance
- Updates FITS headers with corrected focal length and REDUCER field

Setup: Install ASTAP from <https://www.hnsky.org/astap.htm>

4.3.4 Weather Data

(Optional)

Fetches historical weather conditions via Open-Meteo API (free, no API key):

- Temperature, cloud cover, precipitation, wind speed
- Automatic caching (365 days) in SQLite for offline access
- Identifies best observing sessions
- Requires observatory location in configuration

4.4 Output Formats

4.4.1 LaTeX/PDF Report

Professional report with:

- Title page with summary statistics
- Global exposure breakdown by filter
- Per-target sections with detailed statistics
- Observation timeline
- Equipment usage summary
- Automatic fallback to ReportLab PDF if LaTeX is unavailable

4.4.2 HTML Report

Interactive web-based report with:

- Dark cosmic theme matching the application
- Expandable target sections
- Search functionality
- SIMBAD links for each target
- Sortable tables
- Mobile-responsive design

4.4.3 CSV Export

Two CSV files generated:

1. **global_summary.csv**: Overall statistics
2. **targets_summary.csv**: Per-target breakdown

4.4.4 AstroBin CSV

Formatted for direct import into AstroBin with:

- Per-row acquisition parameters (not aggregated per setup)
- Columns: `date`, `filter`, `number`, `duration`, `binning`, `gain`, `sensorCooling`
- Automatic filter name-to-AstroBin ID mapping:
 - Ha → 4663, OIII → 4752, SII → 4844
 - L → 2906, R → 4649, G → 4643, B → 4637
 - NII → 4846
- Per-equipment and per-target export files
- Includes `README.txt` with column descriptions

5 Compression Tab

5.1 Overview

The Compression tab enables bidirectional format conversion with multiple compression algorithms, saving 40–60% disk space while maintaining PixInsight compatibility. Uses parallel processing for maximum speed.

5.2 Basic Workflow

1. **Select Source Folder:** Your uncompressed FITS files
2. **Select Target Folder:** Where compressed files will be saved
3. **Choose Compression Profile:** Select from 9 available profiles
4. **Configure Options:** Delete source, verify integrity, quarantine failures
5. **Preview:** Review estimated space savings
6. **Start Compression:** Click **Start Compression** button

5.3 Compression Profiles

Profile	Ratio	Description
zlib_1	~30%	Minimal compression, fastest. Good for temporary working files.
zlib_6	~50%	Recommended. Best balance of speed and compression. Universal compatibility.
zlib_9	~55%	Maximum zlib compression. Slower, marginal gain over level 6.
zstd_3	~45%	Fast modern compression. Good for quick archival.
zstd_6	~55%	Balanced zstd. Better than zlib_6 with similar speed.
zstd_10	~60%	High compression. Slower but excellent ratio.
zstd_19	~65%	Maximum compression. Very slow. Best for long-term archival.
lz4	~25%	Ultra-fast. Great for working files you access frequently.
lz4_hc	~35%	LZ4 high compression. Still fast, better ratio.
none	0%	No compression. For creating uncompressed XISF or converting FZ→FITS.

Table 2: Compression Profiles

Recommendation: Use `zlib_6` for general archival, `lz4` for working files.

5.4 Supported Conversions

- FITS → XISF (with any compression profile)
- XISF → FITS (decompression)
- FITS → FITS.FZ (RICE tile compression via Astropy)
- FITS.FZ → FITS (decompression via Astropy)
- XISF → FITS.FZ (via FITS intermediate)
- FITS.FZ → XISF (via FITS intermediate)
- XISF → XISF (recompression with different profile)

FITS.FZ Note

FITS.FZ (fpack) compression uses RICE tile compression, handled natively by Astropy. No external `fpack`/`funpack` binaries are needed.

5.5 Options

5.5.1 Delete Source After Compression

WARNING

Enabling this option will **permanently delete** source files after successful compression. Ensure you have backups before using this option!

5.5.2 Verify Integrity

Recommended. Uses SHA-256 checksums to verify:

- File was compressed correctly
- No data corruption occurred
- Decompressed data matches original

5.5.3 Quarantine Failed Files

Moves files that fail compression to a separate quarantine folder for manual inspection rather than leaving them mixed with successful conversions.

5.6 Performance

Compression uses `ProcessPoolExecutor` for parallel processing across all CPU cores:

- **CPU:** More cores = faster (each file compressed in parallel)
- **Storage:** SSD significantly faster than HDD
- **Profile:** lz4 fastest, zstd_19 slowest
- **Auto-tuning:** Worker count automatically adjusted based on system capabilities

6 Header Editor Tab

6.1 Overview

The Header Editor tab enables mass editing of FITS headers across thousands of files with NINA compatibility and automatic pattern building.

6.2 Basic Workflow

1. **Select Files:** Choose files or folder to edit
2. **Review Headers:** Table shows current values organized by category
3. **Edit Values:** Enter new values in “New Value” column
4. **Preview Changes:** Click **Preview** to see what will change
5. **Apply Changes:** Click **Apply to N Files** to write
6. **Undo if Needed:** Use **Undo Last** if you made a mistake

6.3 Header Categories

Headers are organized into categories for easy navigation:

- **Acquisition:** DATE-OBS, EXPTIME, EXPOSURE, IMAGETYP
- **Camera:** INSTRUME, GAIN, OFFSET, READOUTM, EGAIN
- **Image:** OBJECT, OBJECTRA, OBJECTDEC, AIRMASS
- **Filter:** FILTER, FWHEEL
- **Telescope:** TELESCOP, FOCALLEN, APTDIA, APTAREA
- **Focus:** FOCUSPOS, FOCUSTEM, FOCUSNAM
- **Rotator:** ROTATANG, ROTATOR
- **Weather:** PRESSURE, HUMIDITY, AMBTEMP, SKYTEMP

6.4 NINA Pattern Builder

Automatically generates filenames using NINA-compatible tokens:

Available Tokens:

- **\$IMAGETYPE\$** — Light, Dark, Flat, Bias
- **\$TARGETNAME\$** — Object name
- **\$DATETIME\$** — Date and time
- **\$FILTER\$** — Filter name
- **\$BINNING\$** — Binning (e.g., 1x1, 2x2)
- **\$EXPOSURETIME\$** — Exposure in seconds
- **\$ROTATORANGLE\$** — Rotator angle in degrees
- **\$SENSORTEMP\$** — Sensor temperature

- \$TELESCOPE\$ — Telescope name
- \$CAMERA\$ — Camera name

Example:

```
$IMAGETYPE$_$TARGETNAME$_$FILTER$_$EXPOSURETIME$s_$_BINNING$
-> Light_M31_Ha_300s_1x1_0001.fits
```

6.5 Bulk Operations

6.5.1 Copy From Another File

Copy header values from a reference file to selected files:

1. Select target files
2. Click **Copy From...**
3. Choose reference file
4. Select which fields to copy
5. Apply

6.5.2 Add New Field

Add custom FITS header fields:

1. Click **Add Field...**
2. Enter field name (8 characters max, FITS standard)
3. Enter value
4. Optionally add comment
5. Apply to selected files

7 Flat Manager Tab

7.1 Overview

The Flat Manager tab organizes flat frames by characteristics and tracks master flats with automatic coverage reporting.

7.2 Basic Workflow

1. **Scan Library:** Click **Scan** and select your flats folder
2. **Review Groups:** Table shows flats grouped by date/setup/filter/etc.
3. **Check Coverage:** Coverage report highlights missing flats
4. **Create Master Flats:** Select group, click **Create Master Flat**
5. **Link to Targets:** Associate flats with light frames

7.3 Flat Grouping

Flats are automatically grouped by:

- **Date:** Observation date
- **Setup:** Telescope + Camera combination
- **Filter:** Filter name
- **Binning:** Pixel binning (1x1, 2x2, etc.)
- **Rotation:** Rotator angle (for camera rotation)
- **Temperature:** Sensor temperature ($\pm 2^\circ\text{C}$ tolerance)

7.4 Coverage Reports

AstroManager analyzes flat coverage and provides recommendations:

- **Complete** ($\square$): 25+ frames, master flat exists
- **Low Count** ($\triangle$): < 25 frames (recommend acquiring more)
- **Missing** ($\times$): No flats for this setup/filter combination

7.5 Master Flat Tracking

Track which master flats were used for each imaging session:

- Links light frames to appropriate master flat
- Warns if light frame has no matching flat
- Identifies flats used across multiple targets
- Exports flat usage report

8 Target Tracking Tab

8.1 Overview

The Target Tracking tab provides historical observation timelines with integration time statistics, session quality metrics, weather forecasts, and target visibility planning.

8.2 Basic Workflow

1. **Select Target:** Choose from dropdown (auto-populated from database)
2. **View Timeline:** Graphical representation of all sessions
3. **Review Statistics:** Total time, filters, equipment, HFR
4. **Check Best Sessions:** Weather and quality metrics
5. **Plan Ahead:** Weather forecast and target visibility for upcoming nights

8.3 Timeline Visualization

Horizontal bar chart showing:

- **Date:** Observation date (month/year)
- **Duration:** Total integration time (hours)
- **Filters:** Color-coded breakdown by filter
- **Sessions:** Number of distinct imaging sessions

8.4 Statistics Display

Total Integration Time:

- Total hours across all sessions
- Breakdown by filter (Ha, OIII, SII, LRGB, etc.)
- Percentage of total per filter

Equipment:

- Telescopes used (with alias merging)
- Cameras used
- Most common setup

Quality Metrics:

- Best HFR (arc-seconds)
- Best FWHM
- Average values

8.5 Weather Forecast

Plan your next imaging sessions:

- Multi-day weather forecast for your observatory location
- Temperature, cloud cover, humidity, wind speed
- “Imageable night” indicator

8.6 Target Visibility

Check when targets are visible from your location:

- Altitude chart for selected target over the night
- Best imaging window (above 30° altitude)
- Moon phase and distance

8.7 SIMBAD Integration

If SIMBAD data is available:

- Canonical object name
- Object type (Galaxy, Nebula, Star Cluster, etc.)
- Coordinates (RA/DEC)
- Link to SIMBAD entry

9 History & Statistics Tab

9.1 Overview

The History & Statistics tab provides a comprehensive dashboard of your entire astrophotography career. It aggregates data from all analysis runs into a unified view with 7 sub-tabs.

9.2 Dashboard

Six stat cards provide an instant overview:

- Total targets observed
- Total integration time (hours)
- Total frames captured
- Total observation nights
- Average HFR across all sessions
- Number of telescopes used

9.3 Target Rankings

Targets sorted by total integration time, showing:

- Target name and SIMBAD object type
- Total integration time per target
- Number of sessions and filters used
- Telescope(s) used
- Bar chart visualization

9.4 Filter Statistics

Per-filter analysis:

- Usage across targets and sessions
- Total frames and integration time per filter
- Average HFR per filter
- Filter names displayed with Greek Unicode ($H\alpha$, O III, S II)

9.5 Equipment Statistics

Three sub-views:

- **Per-telescope:** Targets, sessions, total integration, average HFR
- **Per-camera:** Same metrics per camera sensor
- **Per-setup:** Statistics for each telescope+camera combination

9.6 Temporal Statistics

Activity breakdowns:

- Monthly activity (sessions and integration time)
- Yearly totals
- Day-of-week distribution

9.7 Best Nights

Top observation nights ranked by:

- Quality (best average HFR)
- Productivity (most integration time)

9.8 Object Type Statistics

Integration time broken down by astronomical object type:

- Emission nebulae, planetary nebulae, galaxies
- Star clusters (open, globular)
- Supernova remnants, HII regions, etc.

9.9 Export and Import

- **JSON Export:** Complete observation history with all metadata
- **CSV Export:** Spreadsheet-compatible format for analysis
- **JSON Import:** Restore from backup or merge from another installation

9.10 Auto-Maintenance

On each refresh, the history tab automatically:

- Merges duplicate targets (same canonical name)
- Normalizes filter names (Greek characters, corrupted entries)
- Normalizes equipment names (removes formatting artifacts)

10 Database Browser Tab

10.1 Overview

The Database Browser tab provides searchable access to AstroManager's built-in databases of astronomical equipment and targets.

10.2 Camera Database

Over 1,600 camera sensors with detailed specifications:

- Pixel size, sensor dimensions, resolution
- Read noise, full well capacity, quantum efficiency
- Bit depth, cooling type, interface
- Filter by brand (ZWO, QHY, Atik, FLI, SBIG, Player One, etc.)

10.3 Telescope Database

Over 3,900 telescope models:

- Aperture, focal length, f-ratio
- Optical design (Newtonian, Schmidt-Cassegrain, APO refractor, etc.)
- Filter by brand (Takahashi, Celestron, Sky-Watcher, Meade, etc.)

10.4 Filter Database

Over 1,600 astronomical filters:

- Central wavelength, bandwidth, transmission
- Filter type (narrowband, broadband, LRGB, photometric)
- Filter by brand (Baader, Astronomik, Chroma, ZWO, Optolong, etc.)
- Greek Unicode display ($H\alpha$, $H\beta$, O III, S II, N II)

10.5 Target Database

Over 32,900 astronomical objects:

- Messier (110), NGC/IC (13,000+), Sharpless, Barnard, Caldwell
- Arp, RCW, LBN, PGC, Abell, LDN, Hickson, UGC
- Coordinates (RA/DEC), object type, common names
- Search by catalog designation or common name

11 Disk Space Tab

11.1 Overview

The Disk Space tab analyzes storage usage, provides actionable optimization recommendations, and includes a file organization feature.

11.2 Storage Analysis

Format Breakdown:

- **FITS (uncompressed):** Size and percentage
- **XISF:** Compressed with profile breakdown
- **FITS.FZ:** fpack compressed
- **Duplicates:** Wasted space from duplicate files

Graphical Visualization:

- Horizontal bar chart showing relative sizes
- Color-coded by format
- Duplicate overlay in red

11.3 Optimization Recommendations

AstroManager analyzes your library and suggests optimizations:

11.3.1 Compress Uncompressed FITS

Example:

- **Current:** 876 GB uncompressed FITS
- **Profile:** zlib_6 (50% compression)
- **Result:** 438 GB compressed
- **Savings:** 438 GB

11.3.2 Remove Duplicates

Example:

- **Duplicates Found:** 1,243 files
 - **Wasted Space:** 148 GB
 - **Types:** Name-based (432), Content-based (811)
- Action:** Click **Extract Duplicates** to move to separate folder.

11.4 File Organization

The Disk Space tab includes a file organization feature with presets:

- **By Type:** Lights, Darks, Flats, Bias into separate folders
- **By Date:** YYYY-MM-DD folder structure
- **By Target:** Group files by object name
- **Custom:** Define your own organization rules

11.5 Storage Type Detection

AstroManager automatically detects storage type:

- **SSD (NVMe/SATA):** Fast random access, optimal for working files
- **HDD (7200rpm/5400rpm):** Slower sequential access, good for archives

Detection methods:

- **Windows:** WMI/PowerShell disk queries
- **Linux:** `/sys/block/<device>/queue/rotational` via `df`
- **macOS:** `diskutil info`

12 Settings Dialog

Access via **Tools** → **Settings** or **Ctrl+,**

12.1 General

- **Language:** Auto-detect, English, or Français

12.2 Observatory

- **Latitude:** Decimal degrees (−90 to 90)
- **Longitude:** Decimal degrees (−180 to 180)
- **Elevation:** Meters above sea level (0–9000m)
- **Timezone:** IANA timezone (e.g., Europe/Paris, America/New_York)

Critical for weather data and target visibility! Update with your observatory location.

12.3 Performance

- **Workers:** Number of parallel workers (0 = auto-detect from CPU/RAM)
- **Batch size:** Files per batch (50–10000, higher = faster but more RAM)

12.4 Compression Defaults

- **Default profile:** Compression profile for new operations
- **Delete source:** Whether to delete originals after compression
- **Verify integrity:** SHA-256 verification after compression

12.5 Analysis Defaults

- **SIMBAD:** Enable/disable by default
- **Plate solving:** Enable/disable by default
- **Weather:** Enable/disable by default

12.6 Bug Reporting

- **Enable:** Send anonymous crash reports (OS, Python version, stack trace — no personal data)

13 Configuration File

13.1 Location

~/.astromanager/config.yaml

All settings from the Settings dialog are also editable directly in this YAML file.

13.2 Full Configuration Reference

```
application:
  version: "1.0.0"
  language: "auto" # auto, en, fr
  theme: "cosmic_dark"

system:
  workers: 0 # 0 = auto-detect
  batch_size: 1000 # files per batch
  cache_size: 5000
  enable_caching: true

observatory:
  latitude: 51.4769 # your latitude
  longitude: -0.0005 # your longitude
  elevation_m: 46
  timezone: "UTC"

analysis:
  enable_simbad: true
  enable_plate_solving: false
  enable_weather_fetch: false
  duplicate_detection: true
  generate_thumbnails: false
  generate_graphs: true
  generate_latex: true
  generate_csv: true

compression:
  default_profile: "zlib_6"
  delete_source: false
  verify_integrity: true
  quarantine_failed: true

plate_solving:
  solver: "astap"
  astap_path: null # auto-detect
  astrometry_path: null
  timeout_sec: 60

weather:
  api_provider: "open-meteo"
  api_key: null
  cache_duration_days: 365
```

```
bug_reporting:
  enabled: true
  endpoint_url: "https://bugs.astromanager.io/report"
```

13.3 Command-Line Options

```
python astromanager.py                # Launch GUI
python astromanager.py --help          # Show help
python astromanager.py --version       # Show version
python astromanager.py --reset-config  # Reset config
python astromanager.py --vacuum-db     # Optimize database
python astromanager.py --debug         # Debug logging

# CLI headless analysis
python astromanager.py --folder /path/to/fits
python astromanager.py --folder /path --resolve-simbad --export-csv
python astromanager.py --folder /path --workers 8
python astromanager.py --folder /path --optimize-storage /backup
```

14 Advanced Topics

14.1 Database Management

14.1.1 Location

~/.astromanager/astromanager.db

14.1.2 Tables

- **targets** — Target information (name, coordinates, SIMBAD data)
- **observations** — Observation records (date, filter, exposure, etc.)
- **weather_cache** — Historical weather data (365-day cache)
- **header_cache** — FITS header cache for performance
- **analysis_results** — Saved analysis results

14.1.3 Backup and Optimization

- Automatic backup on each launch
- Manual optimization: Tools → Optimize Database
- CLI: `python astromanager.py -vacuum-db`
- Cache cleanup: Tools → Clear Old Cache (removes entries > 90 days)

14.2 Performance Tuning

14.2.1 For Large Libraries (100k+ files)

```
system:
  workers: 0          # auto-detect
  batch_size: 5000    # if 16GB+ RAM
  cache_size: 10000
```

14.2.2 For Slower Systems

```
system:
  workers: 2
  batch_size: 500
analysis:
  enable_simbad: false
  enable_weather_fetch: false
```

14.2.3 Performance Architecture

- **Parallel file reading:** ThreadPoolExecutor for I/O-bound operations (header reading, dedup signatures)
- **Parallel compression:** ProcessPoolExecutor for CPU-bound operations
- **Auto-tuning:** Worker count and batch size adjusted based on CPU cores, RAM, and storage type
- **Multi-tier caching:** Memory (L1, fast) → SQLite (L2, persistent) → Disk (L3, archival)

14.3 Plate Solving Setup

14.3.1 ASTAP (Recommended)

1. Download from: <https://www.hnsky.org/astap.htm>
2. Install to default location
3. AstroManager auto-detects common paths:
 - Windows: C:\Program Files\astap\astap_cli.exe
 - Linux: /usr/bin/astap, /opt/astap/astap
4. OR set path in Settings or config.yaml

14.3.2 Astrometry.net

1. Install astrometry.net package
2. Download index files for your field size
3. Set solver to “astrometry” in Settings

15 Troubleshooting

15.1 Common Issues

15.1.1 No FITS files found

Symptoms: Analysis returns “0 files found”

Solutions:

1. Check folder path is correct
2. Ensure files have recognized extensions: `.fits`, `.fit`, `.fz`, `.xisf`, `.xifs`, `.xif`
3. Check file permissions (read access required)
4. Try running as administrator (Windows) or with `sudo` (Linux)

15.1.2 Database locked error

Solutions:

1. Close any other instances of AstroManager
2. Close any SQLite browser tools
3. Restart AstroManager
4. If persistent: delete database (it will recreate):

```
rm ~/.astromanager/astromanager.db  
python astromanager.py
```

15.1.3 Slow performance

Solutions:

1. Check storage type (HDD much slower than SSD)
2. Reduce worker count in Settings for HDD storage
3. Disable online features (SIMBAD, weather)
4. Clear old cache via **Tools** → **Clear Old Cache**
5. Increase batch size if you have sufficient RAM (16GB+)

15.1.4 Out of memory error

Solutions:

1. Reduce batch size in Settings (e.g., 500)
2. Reduce cache size in `config.yaml`
3. Close other applications to free RAM

15.1.5 LaTeX compilation fails

Solutions:

1. Install LaTeX (the launcher `run.bat`/`run.sh` offers to do this)
2. The `.tex` source file is always generated — compile it online via Overleaf
3. AstroManager automatically falls back to ReportLab PDF if LaTeX is unavailable
4. HTML report is always generated as an alternative

15.1.6 `os.startfile` error on Linux

This is handled automatically. AstroManager uses `xdg-open` on Linux and `open` on macOS as fallbacks.

15.2 Getting Help

Built-in Help: Press F1 or **Help** → **User Guide**

GitHub Issues: Report bugs and feature requests on the project repository.

When reporting issues, please include:

- AstroManager version (**Help** → **System Info**)
- Operating system and version
- Python version
- Error message (full traceback from console)
- Steps to reproduce

16 Appendix

16.1 Keyboard Shortcuts

Shortcut	Action
Ctrl+,	Open Settings dialog
Ctrl+Q	Quit application
Ctrl+‘	Toggle console panel
F1	Open User Guide

Table 3: Keyboard Shortcuts

16.2 Supported File Extensions

16.3 Project Structure

```

astromanager/
  astromanager.py      # Main launcher
  fits_analyser_gui.py # Core analysis engine
  requirements.txt     # Python dependencies
  run.bat              # Windows launcher
  run.sh               # Linux/macOS launcher

```

Extension	Description
.fits	Standard FITS format
.fit	Alternative FITS extension
.fz	fpack compressed FITS
.fits.fz	fpack compressed FITS (explicit)
.xisf	XISF format (PixInsight native)
.xifs	Alternative XISF extension
.xif	Alternative XISF extension

Table 4: Supported File Extensions

```

core/
  config.py           # Configuration manager
  database.py         # SQLite database
  workers.py          # Unified worker architecture
  signals.py          # Signal bus
gui/
  main_window.py      # Main window
  theme.py             # Cosmic theme engine
  tabs/
    analysis_tab.py   # Analysis tab
    compression_tab.py # Compression tab
    header_editor_tab.py # Header editor tab
    flat_manager_tab.py # Flat manager tab
    target_tracking_tab.py # Target tracking tab
    disk_space_tab.py # Disk space tab
modules/
  compression.py      # Compression engine
  header_editor.py     # Header editing engine
  flat_manager.py      # Flat management engine
  plate_solving.py     # Plate solving wrapper
  weather_api.py       # Weather API client
  bug_reporter.py      # Anonymous crash reporting
  file_organizer.py    # File organization
database/
  cameras.py          # Camera database
  telescopes.py       # Telescope database
  filters.py          # Filter database
  targets.py          # Target database

```

16.4 Credits

AstroManager is built with these excellent libraries:

- **Astropy** — FITS file handling and RICE compression
- **PyQt6** — GUI framework
- **NumPy, Pandas** — Data processing
- **Matplotlib** — Graph generation
- **xisf** — XISF format support
- **Zstandard, LZ4** — Compression algorithms

- **ReportLab** — PDF generation (LaTeX fallback)
- **SQLite** — Database engine
- **PyYAML** — Configuration
- **psutil** — System capability detection
- **scipy** — Advanced statistics
- **astroquery** — SIMBAD integration
- **tqdm** — Progress bars
- **requests** — HTTP requests (weather, bug reports)

Special Thanks:

- SIMBAD (CDS, Strasbourg) for astronomical database
- Open-Meteo for free weather API
- Han Kleijn for ASTAP
- Astrometry.net team
- PixInsight for XISF specification
- CloudyNights community for testing and feedback

16.5 License

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